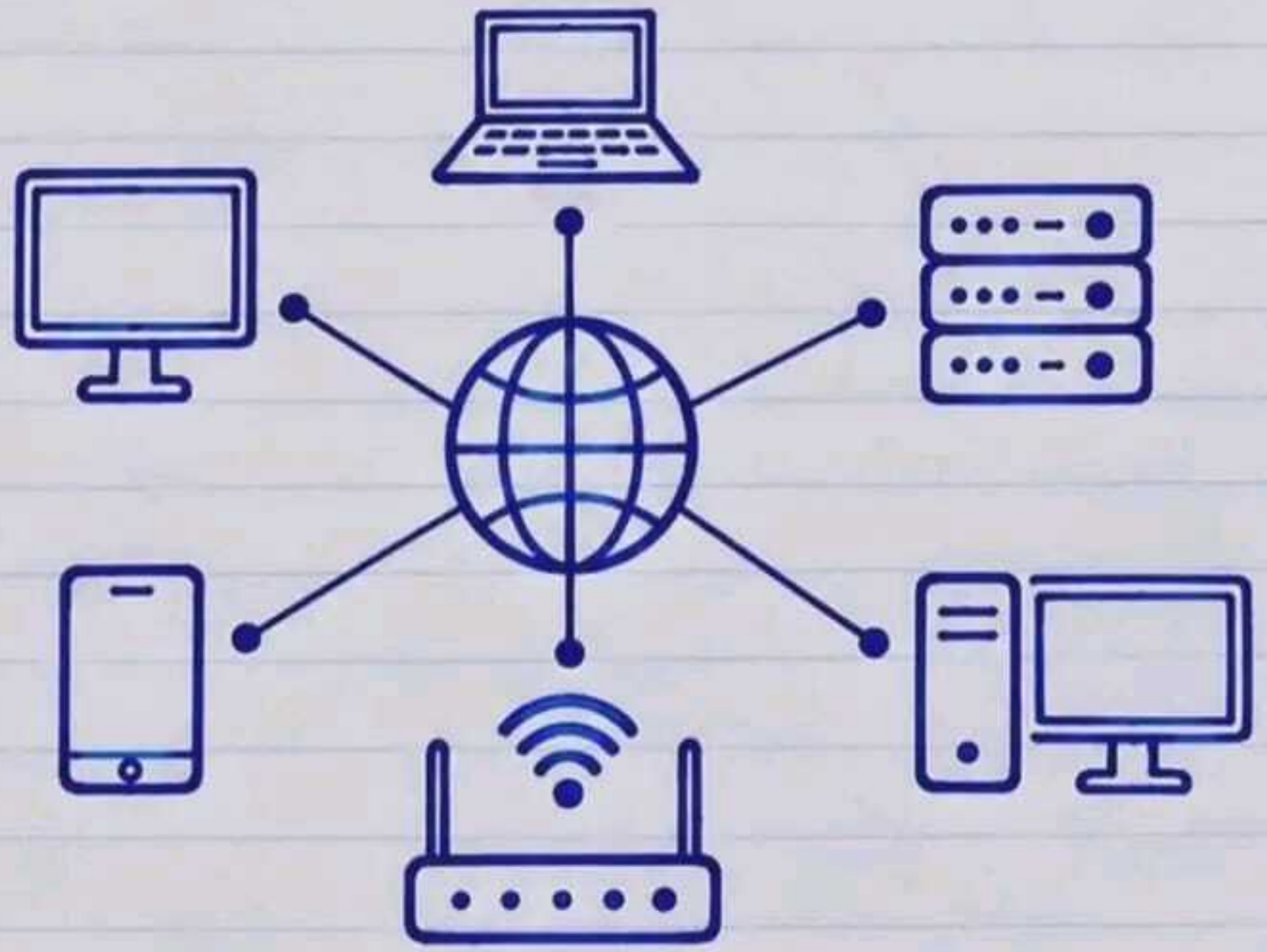
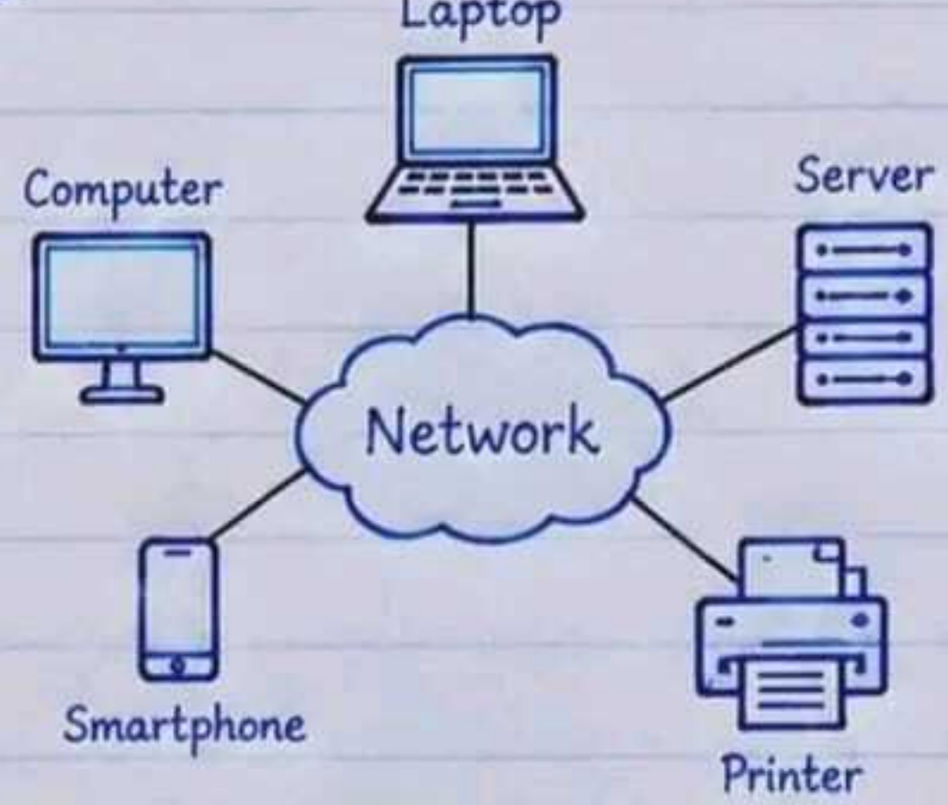
Computer Networks



1. Introduction to Computer Networks

What is a Computer Network?

- A computer network is a group of two or more computers and other devices connected together to share resources and information.



Main Purpose

- Share data and information
- Share hardware and software resources
- Communication between users
- Improve reliability and efficiency

Types of Networks

① **PAN** (Personal Area Network) → Connects devices within a small area (e.g. mobile, smartwatch).

② **LAN** (Local Area Network) → Connects devices in a limited area (e.g. home, office, campus).

③ **MAN** (Metropolitan Area Network) → Covers a city or large area.

④ **WAN** (Wide Area Network) → Covers a country or the whole world.

Conclusion

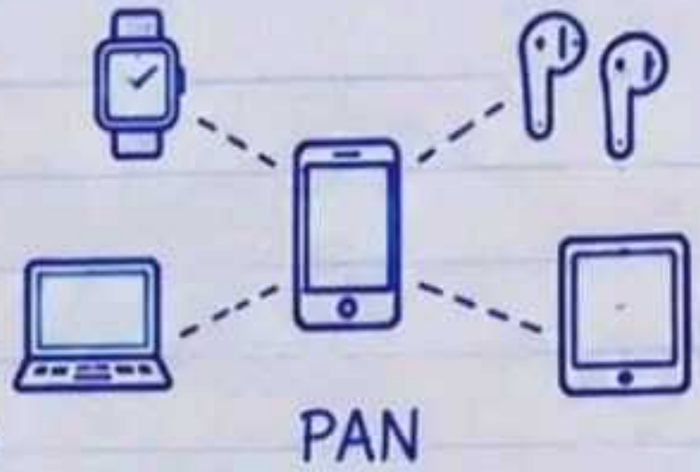
- Computer networks help in connecting people, devices and resources, making communication and data sharing easy and fast.

——— ♥ ———

2. Types of Networks

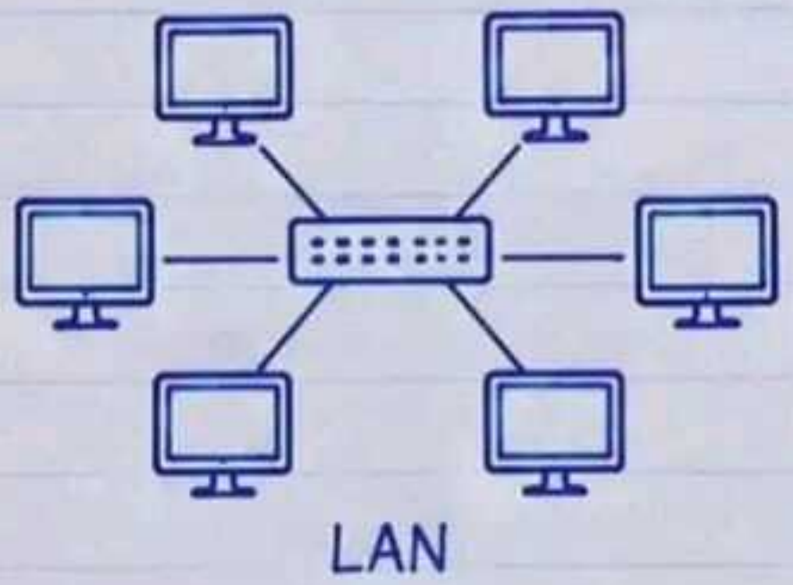
① PAN (Personal Area Network)

- Connects devices within a very small area (e.g. a person's room or around a person).
- Used for short-range communication between personal devices.
- Examples: Bluetooth, mobile hotspot, smartwatch, wireless earphones.



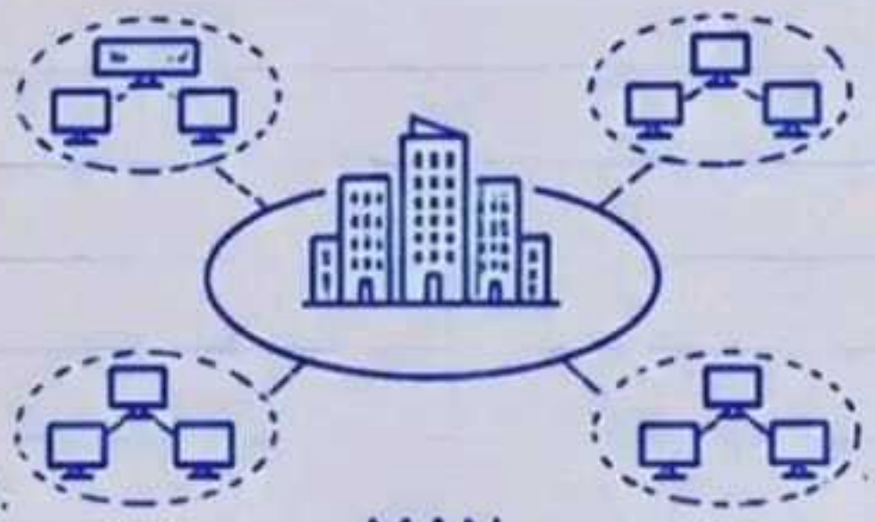
② LAN (Local Area Network)

- Connects devices in a limited area (e.g. home, office, school, campus).
- High speed and reliable.
- Examples: Home network, office network, college lab network.



③ MAN (Metropolitan Area Network)

- Covers a city or large urban area.
- Connects multiple LANs together.
- Usually provided by telecom companies or organizations.
- Example: City-wide network, university network.



④ WAN (Wide Area Network)

- Covers a country or the whole world.
- Connects multiple MANs and LANs across long distances.
- Uses public or private networks (e.g. Internet).
- Example: Internet, global company network.



3. Network Topologies

What is Network Topology?

- Network topology refers to the physical or logical arrangement of devices (nodes) in a network and the way they are connected to each other.

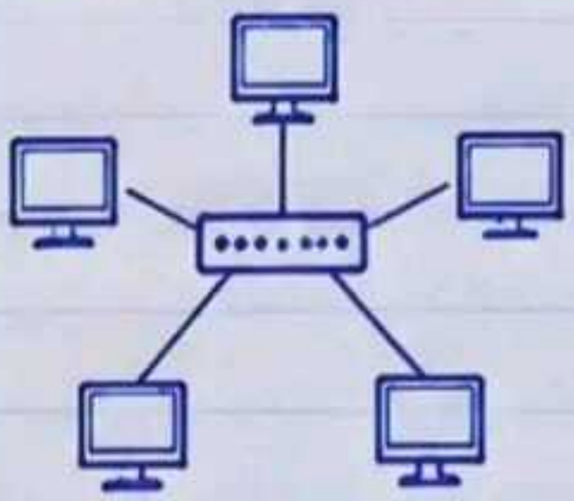
Types of Network Topologies

1 Bus Topology



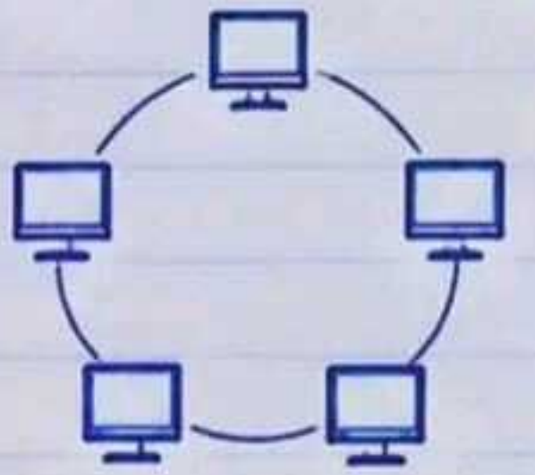
- All devices are connected to a single backbone cable.
- Simple and low cost.
- If the main cable fails, whole network stops.

2 Star Topology



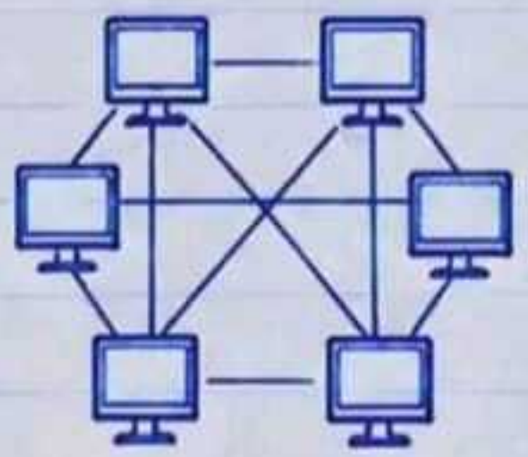
- All devices are connected to a central device (hub/switch).
- Easy to add or remove devices.
- If central device fails, whole network stops.

3 Ring Topology



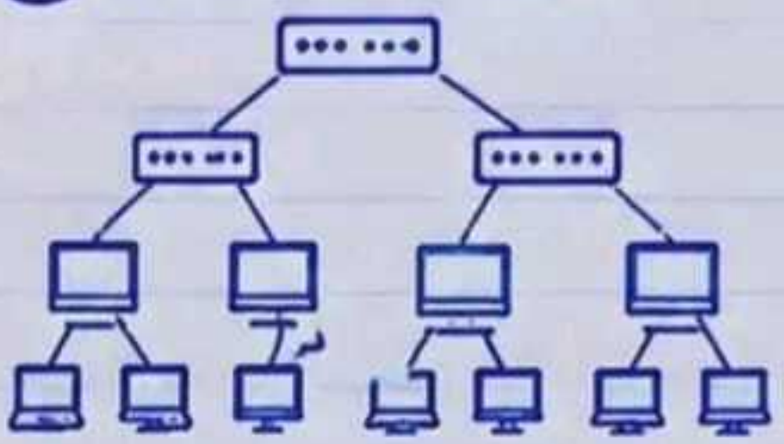
- Devices are connected in a circular form.
- Data moves in one direction (or both).
- If one device fails, whole network can be affected.

4 Mesh Topology



- Devices are connected to multiple devices.
- Provides high reliability and fault tolerance.
- Expensive and complex to set up.

5 Tree Topology



- Combination of star and bus topology.
- Uses a central backbone with multiple star networks.
- Scalable and easy to manage.
- Failure in main backbone affects the entire network.

4. Transmission Media

What is Transmission Media?

- Transmission media are the physical path or channel through which data is transmitted from one device to another in a network.

Types of Transmission Media

1. Guided Media (Wired)

- Signals travel through a physical cable or wire.
- More secure and less affected by external interference.

- **Types :**

a) Twisted Pair Cable

- Two insulated copper wires twisted together.
- Used in telephone and LAN.
- Example: Ethernet cable (UTP/STP).



Twisted pair cable

b) Coaxial Cable

- Central copper wire, insulation and outer shield.
- Provides better speed and less interference than twisted pair.
- Example: Cable TV, broadband.



Coaxial cable

c) Optical Fiber Cable

- Uses light to transmit data.
- Very high speed and high bandwidth.
- Least signal loss, immune to electromagnetic interference.
- Example: Internet, long-distance communication.



Optical fiber cable

2. Unguided Media (Wireless)

- Signals travel through air or space without a physical cable.
- Less secure and more affected by interference.

- **Types :**

a) Radio Waves

- Long range, can penetrate buildings and obstacles.
- Used in radio, TV and mobile networks.



Radio waves

b) Microwave

- High frequency, short range.
- Used for point-to-point communication and satellite.



Microwave

c) Infrared

- Short range, line of sight.
- Used in remote controls and short-range devices.



Infrared

Key Point

- Guided media = cable/wire | Unguided media = wireless (air/space)

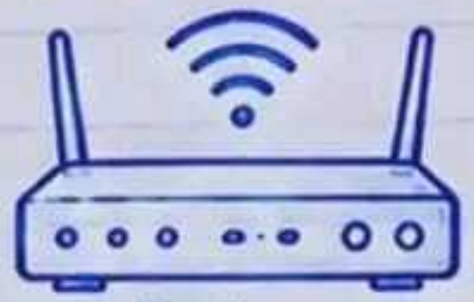
5. Network Devices

What are Network Devices ?

- Network devices are hardware components that connect devices, control the flow of data and help in communication between different networks.

(1) Router

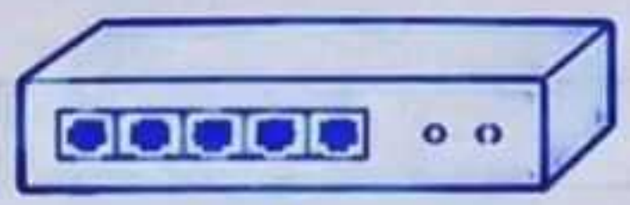
- Connects multiple networks (e.g. LAN to Internet).
- Directs data packets using IP addresses.
- Works at Network layer (Layer 3).
- Example: Home router, edge router.



Router

(2) Switch

- Connects multiple devices in a LAN.
- Forwards data only to the required device using MAC address.
- Works at Data Link layer (Layer 2).
- Example: Ethernet switch.



Switch

(3) Hub

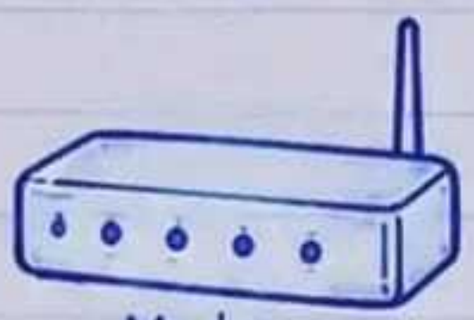
- Connects multiple devices in a LAN.
- Sends data to all devices (no filtering).
- Works at Physical layer (Layer 1).
- Example: Old Ethernet hub.



Hub

(4) Modem

- Connects a local network to the Internet.
- Modulates and demodulates digital and analog signals.
- Works at Physical layer (Layer 1).
- Example: DSL modem, cable modem.



Modem

(5) Firewall

- Monitors and controls incoming and outgoing traffic.
- Protects the network from unauthorized access.
- Works at Network/Transport layer (Layer 3/4).
- Example: Hardware firewall, software firewall.



Firewall

6. IP Addressing & Subnets

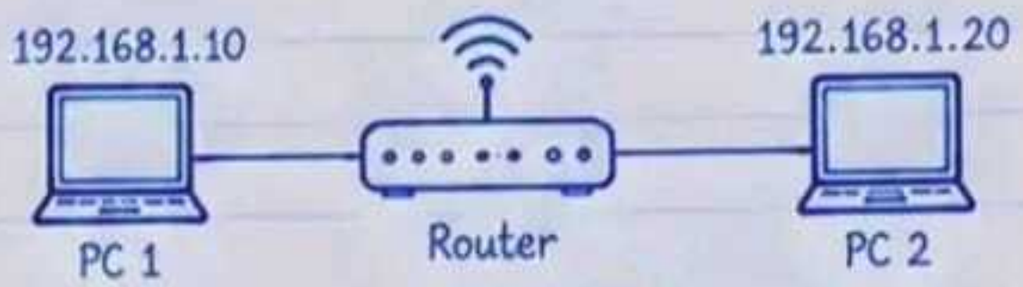
What is IP Address?

- IP (Internet Protocol) address is a unique number given to each device on a network.
- It helps in identifying the device and enables communication between devices.
- Used at Network layer (Layer 3).

Example :

IPv4 address : 192.168.1.10

IPv6 address : 2001:0db8:85a3::8a2e:0370:7334



Types of IP Addresses

① IPv4 Address

- 32 bits (4 bytes).
- Written in dotted decimal form (e.g. 192.168.1.10).
- Provides around 4.3 billion unique addresses.

② IPv6 Address

- 128 bits (16 bytes).
- Written in hexadecimal form (e.g. 2001:0db8:85a3::8a2e:0370:7334).
- Provides a vastly larger number of addresses.

IP Address Classes (IPv4)

- Class A : 1.0.0.0 – 126.0.0.0
- Class B : 128.0.0.0 – 191.255.0.0
- Class C : 192.0.0.0 – 223.255.255.0
- Class D : 224.0.0.0 – 239.255.255.255 (Multicast)
- Class E : 240.0.0.0 – 255.255.255.255 (Experimental)

What is Subnet?

- A subnet is a smaller network created from a larger network.
- It helps in better management of IP addresses, reduces broadcast traffic and improves security.
- Done using subnet mask.

Example of Subnetting

Network : 192.168.1.0/24

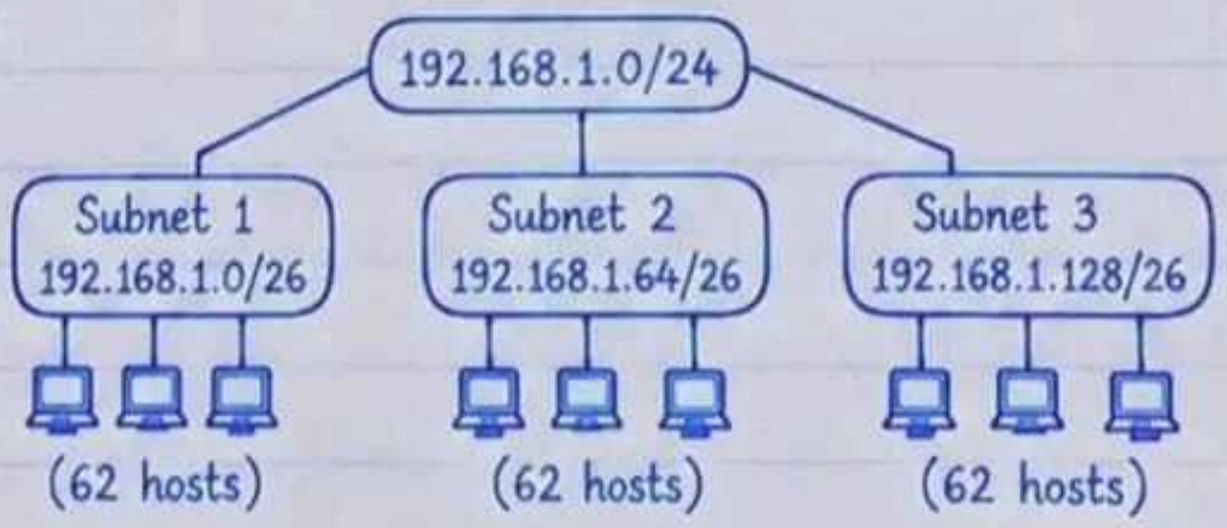
Subnet Mask : 255.255.255.0

Possible Subnets : 256

Usable Hosts per Subnet : 254

Why Use Subnetting?

- Efficient use of IP addresses.
- Reduces network traffic.
- Improves security and performance.
- Helps in network organization.



7. Network Protocols

What are Network Protocols?

- Network protocols are a set of rules and standards that define how data is formatted, transmitted, receivea and processed in a network.
- They ensure proper communication between devices of different types and manufacturers.
- Without protocols, devices cannot understand each other.

Key Points

- Protocols act as a common language for devices.
- They define the format of data, timing, error control and more.
- Follow standards (e.g. by IETF - Internet Engineering Task Force).

Types of Network Protocols

1 TCP/IP Protocol Suite

- Most widely used protocol suite.
- Includes IP, TCP, UDP, HTTP, FTP, etc.
- Used in the Internet and most modern networks.

2 HTTP (Hypertext Transfer Protocol)

- Used for transferring web pages.
- Works on top of TCP/IP.
- Example: Browsing websites.

3 FTP (File Transfer Protocol)

- Used for transferring files between devices.
- Works on top of TCP/IP.
- Example: Uploading or downloading files.

4 SMTP (Simple Mail Transfer Protocol)

- Used for sending emails.
- Works on top of TCP/IP.
- Example: Sending an email from one user to another.

5 DNS (Domain Name System)

- Translates domain names to IP addresses.
- Helps us access websites using names (e.g. google.com).
- Example: www.google.com → 142.250.74.68

TCP/IP Model (Layers)

Application	Provides network services (HTTP, FTP, SMTP, DNS etc.)
Transport	Handles end-to-end delivery (TCP, UDP)
Internet	Handles logical addressing (IP, ICMP, IGMP)
Network Access	Handles physical transmission (Ethernet, Wi-Fi, PPP etc.)

Example of Protocol Communication



Data is broken into packets, given a protocol (like TCP/IP), and sent through the network. At the receiver, the protocol ensures the data is reassembled correctly.

Conclusion

- Network protocols make communication possible between different devices and networks. They are the backbone of the Internet and modern networking.

9. Network Layers & Models

What are Network Layers?

- Network layers are logical levels in a network that perform specific functions to enable communication between devices.
- Each layer works with the layer above and below it.

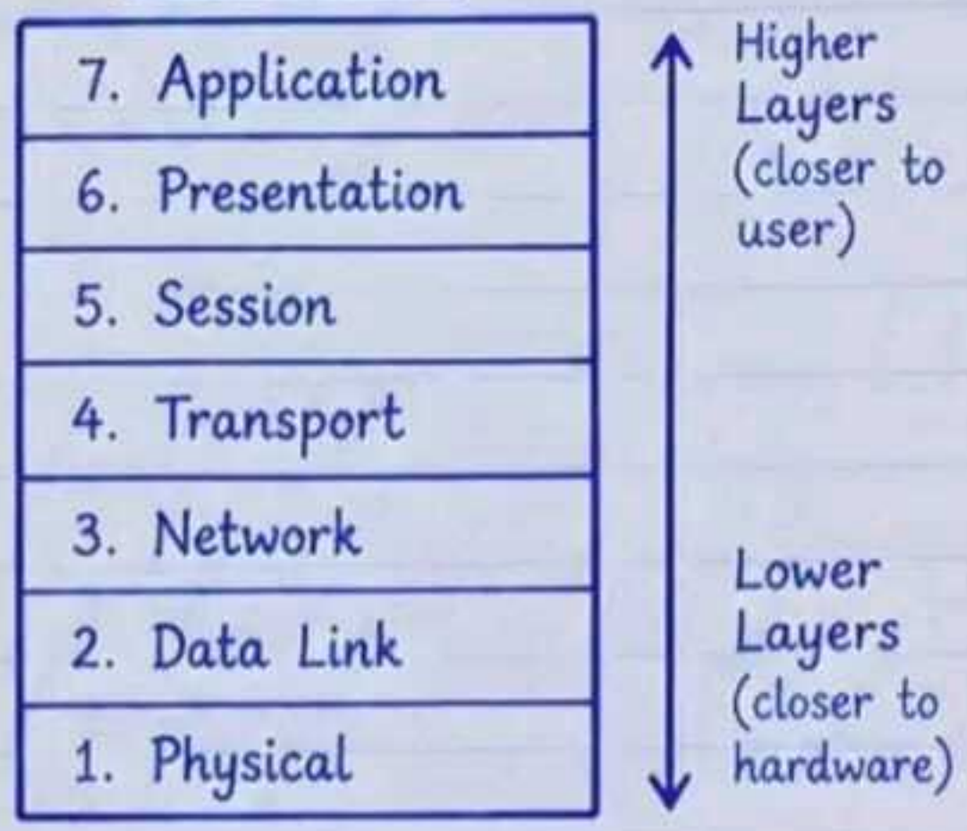
Key Points

- Layers help in standardizing communication.
- Each layer has its own protocols.
- Data is encapsulated as it moves down the layers and de-encapsulated at the receiver side.

The OSI Model (7 Layers)

1. **Physical Layer** – Handles physical transmission of data (bits, signals, cables).
2. **Data Link Layer** – Provides error detection and frame control.
3. **Network Layer** – Handles logical addressing and routing.
4. **Transport Layer** – Provides end-to-end communication (reliability, flow control).
5. **Session Layer** – Manages sessions between applications.
6. **Presentation Layer** – Deals with data format, encryption and compression.
7. **Application Layer** – Provides network services to end users (e.g., HTTP, FTP, SMTP).

OSI Model



The TCP/IP Model (4 Layers)

1. **Application** – Combines OSI application, presentation and session layers.
2. **Transport** – Provides end-to-end communication (TCP/UDP).
3. **Internet** – Handles logical addressing and routing (IP, ICMP).
4. **Network Access** – Handles physical transmission and data link functions (Ethernet, Wi-Fi, etc.).

Comparison

Feature	OSI Model	TCP/IP Model
Layers	7	4
Developed by	ISO	IETF
Focus	Theoretical	Practical
Usage	Less common	Widely used

Example

- When you open a website, data goes through multiple layers of the TCP/IP model to reach the destination.